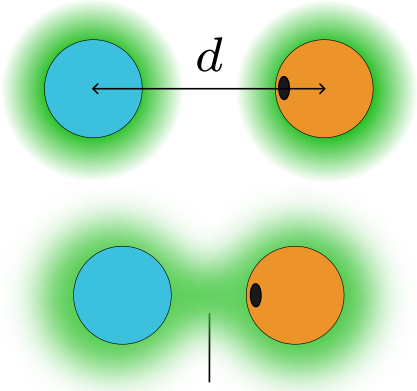
Bulk-mediated attraction



Capillary bridge

Cahn-Hilliard total energy

$$F = \int_B \left[\frac{\kappa}{2}(\nabla\phi)^2 + f(\phi)\right] d^3x - \int_{S_1} h_0\phi(\mathbf{x})dS - \int_{S_2} h_0\phi(\mathbf{x})dS$$

$$U_{\text{attr}}(d) = \min_{\phi(\mathbf{x})} \{F\}$$

$$\text{Force} = \partial_d U_{\text{attr}}(d)$$